Graduation Project Proposal

Development of a Microscopic Traffic Simulation Platform

Overview

The aim of this graduation project is to design and implement a microscopic traffic simulation platform with an initial focus on signalized urban intersections. The simulator will model individual vehicles and their interactions with surrounding traffic and traffic signals, allowing different operating conditions and control strategies to be studied in a controlled environment.

Traffic simulation is commonly used to examine congestion, intersection performance, travel times, delay, queues, and the effect of different signal-control settings. For this project, the main emphasis will be on traffic behavior at and around signalized intersections.

The proposed application will represent the elements needed to simulate selected signalized intersections, including the intersection geometry, traffic movements, signals, and individual vehicles. A user will be able to set up a scenario, run the simulation, observe vehicle movement, and review the resulting traffic measures.

An important part of the project is to represent traffic behavior that is relevant to Egyptian urban conditions. In particular, the simulator should be able to study non-lane-based or weak-lane-discipline traffic, where vehicles may interact laterally and use the available road width more flexibly than in conventional lane-based models.

The project may be developed from scratch, built on an existing open-source traffic-simulation project, or developed from another suitable starting point selected by the students and supervisors. SUMO (Simulation of Urban MObility) is one possible open-source platform that may be studied, extended, or used as a reference. If an open-source code base is used, the students' work and extensions should remain open-source and comply with the applicable license.

Project Objectives

The main objective is to build a microscopic traffic simulator focused initially on signalized intersections. Each vehicle will be represented separately, and its position and behavior will be updated as the simulation progresses, including interactions that do not depend on strict lane discipline.

The work draws on several areas studied in the Computer Department, including:

- Software engineering
- Object-oriented programming
- Algorithms and data structures
- Mathematical modeling

- Computer simulation
- Open-source software development and integration, where applicable
- User-interface design
- Visualization
- Database systems
- Artificial intelligence, where appropriate

By the end of the project, users should be able to define the traffic demand for a selected intersection scenario, configure signal timings, run the simulation, observe individual vehicle movement, and obtain useful traffic-performance measures. The simulator should also support non-lane-based or weak-lane-discipline vehicle interactions.

Scope of Work

1. Signalized Intersection Modeling

Signalized intersections will form the starting point of the simulator. The model should represent the geometry and traffic movements needed for a selected intersection, including:

- Approaches and exits
- Traffic movements through the intersection
- Intersection geometry
- Signal groups and phases
- Green, amber, and red times
- Permitted and conflicting movements
- Stop lines and vehicle discharge through the intersection

2. Vehicle Generation

The simulator will generate vehicles entering the intersection approaches according to configurable traffic-demand values.

Each generated vehicle may have attributes such as:

- Vehicle type and dimensions
- Approach and intended turning movement
- Desired speed
- Entry time
- Acceleration and deceleration characteristics
- Behavioral parameters used by the movement model

Traffic demand should be adjustable so that both normal operation and congested conditions can be tested.

3. Vehicle Movement and Behavior

A central part of the project is the vehicle movement model: how a vehicle moves through the available road space and responds to other vehicles, traffic signals, and intersection conditions.

The model may cover:

- Acceleration and deceleration
- Car-following behavior
- Safe-distance maintenance
- Lateral positioning and gap use
- Stopping at traffic signals
- Intersection crossing
- Reaction to vehicles ahead and to vehicles alongside

The exact traffic-behavior models will be selected during the project based on the required level of realism and the available development time.

. Non-Lane-Based Traffic Behavior

The simulator should not assume that vehicles always follow ideal lane discipline. This is particularly important for Egyptian urban traffic, where non-lane-based or weak-lane-discipline movement is common and vehicles may use the available road width more flexibly. Depending on the selected modeling approach, this may involve:

- Representing lateral position within the available road width
- Allowing side-by-side movement where sufficient space exists

Vehicles may seek and use available gaps rather than remaining centered on fixed lane positions, while still respecting safe movement and intersection-control rules.

5. Traffic Signal Simulation

Traffic signals are a primary component of the project and should be modeled from the beginning. The user should be able to configure items such as:

- Signal phases
- Green, amber, and red durations
- Cycle time
- Phase sequence
- Allowed traffic movements

This will make it possible to run the same traffic demand under different signal settings and compare their effect on delay, queue length, throughput, and other performance measures.

6. Simulation Engine

The core of the application will be the simulation engine. It will advance simulation time and update the state of the vehicles, signals, and other simulation elements at each step.

Its main responsibilities will include:

- Vehicle movement
- Vehicle interactions
- Traffic-signal updates
- Intersection behavior
- Traffic events
- Collection of simulation statistics

This is expected to be one of the main technical challenges of the project, particularly when the number of vehicles increases.

7. Development Approach and Open-Source Option

The students are not restricted to a single implementation approach. They may develop the simulator from scratch or build on a suitable open-source project, subject to agreement with the supervisors.

SUMO is one possible starting point. It may be studied as a reference, used for comparison and validation, or extended where this supports the project objectives. Other open-source starting points may also be considered.

If an existing open-source platform is used, the graduation-project contribution must remain substantial and clearly identifiable. The students should document what they designed or changed, and their work should remain open-source in accordance with the applicable license. Simply configuring an existing simulator without meaningful development will not be sufficient.

8. Visualization and User Interface

A graphical interface will be used to show the simulation while it is running.

At minimum, the display should show:

- Intersection geometry
- Approaches and permitted movements
- Vehicle positions within the available road width
- Traffic signals
- Moving vehicles
- Traffic conditions

The interface should also provide basic simulation controls, including:

- Start
- Pause

- Stop
- Restart
- Change simulation speed
- Configure simulation parameters

9. Traffic Performance Analysis

Running a simulation is only useful if its results can be measured. The application will therefore calculate indicators such as:

- Average vehicle speed
- Travel time
- Vehicle delay
- Queue length
- Waiting time
- Traffic flow
- Intersection throughput
- Vehicles served per signal cycle
- Congestion level

Results should be presented in a form that is easy to review and compare, for example through tables, charts, or summary reports.

10. Scenario Simulation

Users should be able to modify traffic demand, vehicle behavior, or signal-control settings and compare one scenario with another.

Possible scenarios include:

- Increasing or decreasing traffic volume
- Changing traffic-signal timing
- Changing turning-movement demand
- Adding or removing a lane
- Changing the proportion or size of vehicle types
- Changing intersection capacity
- Comparing strict lane-based and non-lane-based behavior
- Simulating an accident or temporary blockage
- Modifying the intersection or signal configuration
- Comparing different signal-control strategies

11. Testing and Validation

Testing will cover both software correctness and the credibility of the simulated traffic behavior.

Where suitable data or reference cases are available, results may be checked against:

- Reference traffic scenarios
- Available real traffic observations
- Analytical calculations
- Results from established microscopic traffic-simulation software such as SUMO

Expected Deliverables

The expected project deliverables are:

1. A functional microscopic traffic simulation software application.
2. A configurable representation of one or more signalized intersection scenarios.
3. Traffic-demand and vehicle-generation functions.
4. Microscopic vehicle movement and interaction models, including non-lane-based or weak-lane-discipline traffic behavior.
5. Traffic-signal simulation.
6. Real-time graphical visualization.
7. Traffic-performance analysis and reporting.
8. Scenario creation and comparison functionality.
9. Testing and validation results.
10. Technical system documentation.
11. Graduation project report.
12. Final presentation and live demonstration.

Possible Advanced Extensions

Once the core simulator is working reliably, the team may consider one or more of the following extensions:

- Adaptive traffic-signal control
- Optimization of signal timings
- Machine-learning-based congestion prediction
- Reinforcement-learning-based traffic-signal control
- Emergency vehicle priority
- Public transportation simulation
- Pedestrian simulation
- Accident and incident modeling
- Integration with real traffic data
- Web-based simulation
- 3D visualization

These are optional. They should only be attempted after the basic microscopic simulation, signalized-intersection behavior, non-lane-based traffic model, visualization, and analysis functions are complete and stable.

Recommended Background

The project is best suited to students who are comfortable with programming and have a reasonable background in some of the following areas:

- Object-oriented programming
- Data Structures and Algorithms
- Software engineering
- Database systems
- Web or desktop application development
- Computer graphics or visualization
- Artificial intelligence or machine learning
- Mathematical modeling and simulation

No previous transportation-engineering background is required. The traffic concepts needed for the project will be introduced as they become relevant, with guidance from the supervisors.

Who Should Consider This Project?

This project would suit students who are interested in:

- Building a complete software system from design to implementation
- Algorithms and problem solving
- Simulation and modeling
- Graphical and interactive applications
- Artificial intelligence and optimization
- Smart cities and intelligent transportation systems
- Applying computer engineering techniques to real-world engineering problems

Expected Project Outcome

The expected outcome is a working prototype of a microscopic traffic simulator focused initially on signalized intersections. The completed system should be able to model and visualize individual vehicle movement, represent non-lane-based traffic behavior, evaluate alternative signal settings, report traffic-performance measures, and support comparisons between different scenarios.

From a computer-engineering perspective, the project gives the team a chance to work on a substantial software system that combines architecture, algorithms, simulation, visualization, and optimization around a real engineering problem.

Expected Team Size

The project is expected to accommodate a team of 8 students.

Teams interested in the project are invited to send the CVs of all team members by email to youssra.elqattan@eng.asu.edu.eg, with the subject line “Traffic Graduation Project.”

Supervisors

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